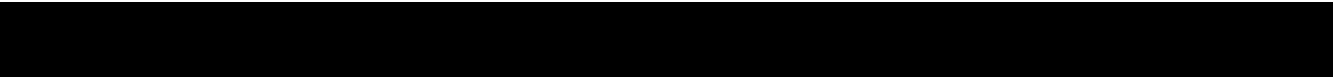


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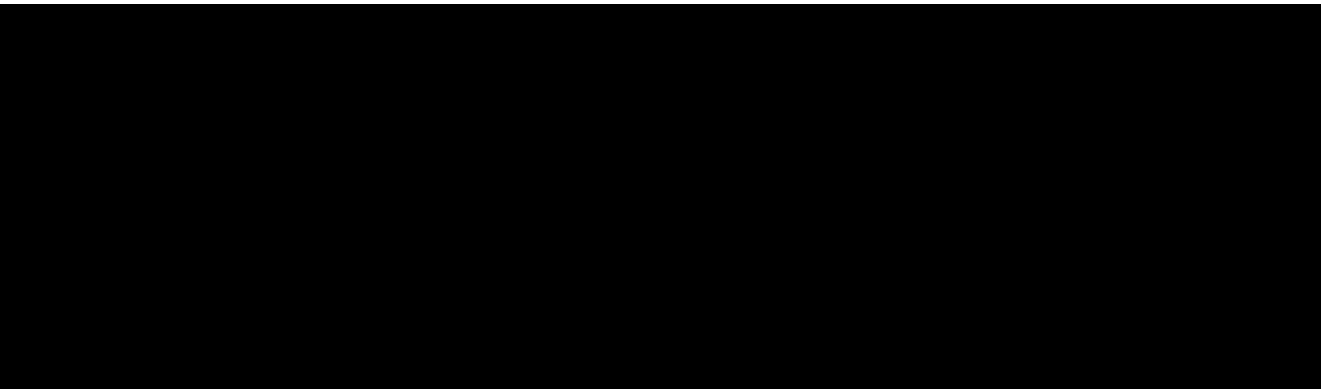


VERA is a fully custom, AI-powered personal operating system built for Ahmed and Victoria. It connects Google Calendar, Google Sheets, Gmail, and Anthropic's Claude AI into a unified system that monitors every domain of life — from PTO burn-down to gym attendance to upcoming birthdays — and surfaces what matters, when it matters.

VERA operates across three interaction modes:

- 1. Proactive Intelligence — nightly AI scans that generate prioritised flags delivered to your inbox at 7 AM
- 2. Conversational Chat — a Claude-powered chat interface where you can ask questions, take actions, and manage your life in plain English
- 3. Dashboard — a React web app with 20+ tabs giving live access to every data domain

Key Numbers at a Glance



Share Backend

2.1 System Architecture

VERA is built on a three-layer architecture that requires no traditional infrastructure — no servers, no databases, no build pipelines.

2.2 Nightly Intelligence Data Flow

The nightly run executes at 11 PM and orchestrates all intelligence jobs in sequence:

1. Read Google Calendar (configurable days ahead)
2. Read all Sheet tabs (tasks, goals, finance, PTO, interests, etc.)
3. Build structured prompt with live context
4. Call Claude API — receive JSON flags array
5. Deduplicate against existing flags (key-based)
6. Write new flags to Flags tab with urgency, reason, key
7. Escalate aged flags (Day 3: bump urgency, Day 7: add stale marker)
8. Run 15+ specialist jobs (Important Dates, Fitness, PTO, Travel, etc.)
9. At 7 AM: send morning email with flag summary grouped by urgency

2.3 Chat Interaction Data Flow

1. User sends message to dashboard
2. Frontend sends message + session ID to backend via GET ?action=chat
3. Backend detects intent (finance / travel / health / people / home / career / goals / resources)
4. Load relevant context from Sheets (only data matching intent)
5. Build system prompt with live data injected
6. Call Claude API with conversation history (last 10 exchanges)
7. Claude returns text + optional ACTION lines
8. Backend executes each ACTION
9. If fetch_resource_content: fetch Google Doc, make second Claude call with content injected
10. Strip ACTION lines, return clean reply to user
11. Save exchange to conversation history

2.4 Authentication & Security

- Single VERA_WEB_TOKEN stored in Script Properties
- All dashboard API calls include ?token=TOKEN
- Token validated on every request — 403 returned on mismatch
- No user accounts, no OAuth flows for end users
- Claude API key stored in Script Properties (never in code)

The VERA dashboard is a React single-page application with 20+ tabs, each providing live access to a specific domain. All data is fetched on-demand via authenticated API calls to the Apps Script web app.

3.1 Home Tab

Bento-card overview surfacing the most important signals across all domains:

- Active flags count (High / Medium / Low)
- Open tasks with overdue indicator
- Active projects count
- Monthly spending chart
- Shopping list summary
- PTO status
- Upcoming milestones countdown
- Active Travel Card (trip-aware weather, live flight status, today's itinerary)

3.2 Chat Tab

Full conversational AI interface powered by Claude. Supports 40+ real executable actions, text and image attachments, and maintains conversation history per session.

3.3 Flags Tab

- View all flags sorted by urgency
- Filter by status (active / acknowledged / snoozed / resolved)
- Acknowledge, snooze (N days), or resolve any flag
- Escalation indicators and flag source shown

3.4 Tasks Tab

- Open tasks with due date, overdue (red), neglected (yellow, 7+ days)
- Add tasks with optional due date and recurring interval
- Complete / edit / delete tasks
- Recurring task auto-creation on completion (Daily / Weekly / Bi-weekly / Monthly / Quarterly / As-needed)

3.5 Projects Tab

- Project list with sub-tasks, priorities (High / Medium / Low), and due dates
- Progress bar per project
- Add / complete / edit / delete via dashboard or chat

3.6 Goals Tab

- Yearly goals in kanban-style view
- Status: Not Started / In Progress / Done / Paused
- Progress percentage tracking
- Category-based organisation

3.7 Shopping Tab

- Multi-store lists (each store is a separate column group)
- Toggle items purchased / unpurchased
- Add items to any store
- Recipe-to-shopping integration

3.8 PTO Planner Tab

- Two subtabs (Ahmed and Victoria)
- Vacation days remaining / used / buffer
- Personal hours tracking
- 3-2-1 Framework: Long Weekends (3–4 workdays) / Mid-Size Weeks (5–7 workdays) / Big Pivot (8+ workdays)
- Clear Windows with one-click Vera Calendar suggestion
- Upcoming milestones and Trigger Buffer

3.9 Travel Tab

- Upcoming trips with departure countdown
- Itinerary editor with 15+ item types (flight, hotel, dining, museum, cruise, spa, ski, etc.)
- Packing list with person attribution and category grouping
- Packing progress percentage
- Live flight status
- Trip context, AI trip recommendations, and traveler profiles

3.10 People Tab

- Important Dates subtab: upcoming birthdays, anniversaries, meaningful dates
- Gift Ideas subtab: braindump of gift ideas per person

3.11 Home Front Tab

- Chores subtab: by cadence (Daily / Weekly / Bi-weekly / Monthly / Quarterly / As-needed), reset on schedule
- Vehicles subtab: oil change interval, registration / insurance expiry, mileage tracking, service history

3.12 Health & Wellness Tab

- Prescriptions subtab: medications for Ahmed and Victoria with dosage, frequency, refill dates
- Gym Log subtab: session history, attendance status, week / month grouping, pending check-ins
- Morning Routine: daily habit checklist that auto-resets nightly

3.13 Home Items Tab

- Appliances and warranties with purchase date and warranty expiry
- Service interval configuration with last / next service date tracking
- Record service (auto-creates Google Calendar reminder)
- Maintenance countdown indicators

3.14 Interests Tab

- Shared interest ledger for Ahmed and Victoria
- Categories: Food / Travel / Fitness / Culture / Hobbies / Learning / Other
- Used by VERA for gift suggestions and event cross-referencing

3.15 Ideas Tab

- Free-form braindump with category and tag support
- Promote to Task (one-click)
- Archive ideas

3.16 Finance Tab

- Bills subtab: monthly / annual bills with due date, amount, paid status
- Credit Cards: cards per person with reward rates, perks, last-used tracking
- Loyalty Programs: points balances, tiers, redemption goals
- Financial Goals: savings goals with target, current, monthly contribution, APY, target date
- What-If Scenarios: simulate purchases or recurring changes
- Transaction History: Empower CSV integration, spending by category, cashflow chart

3.17 Career Tab

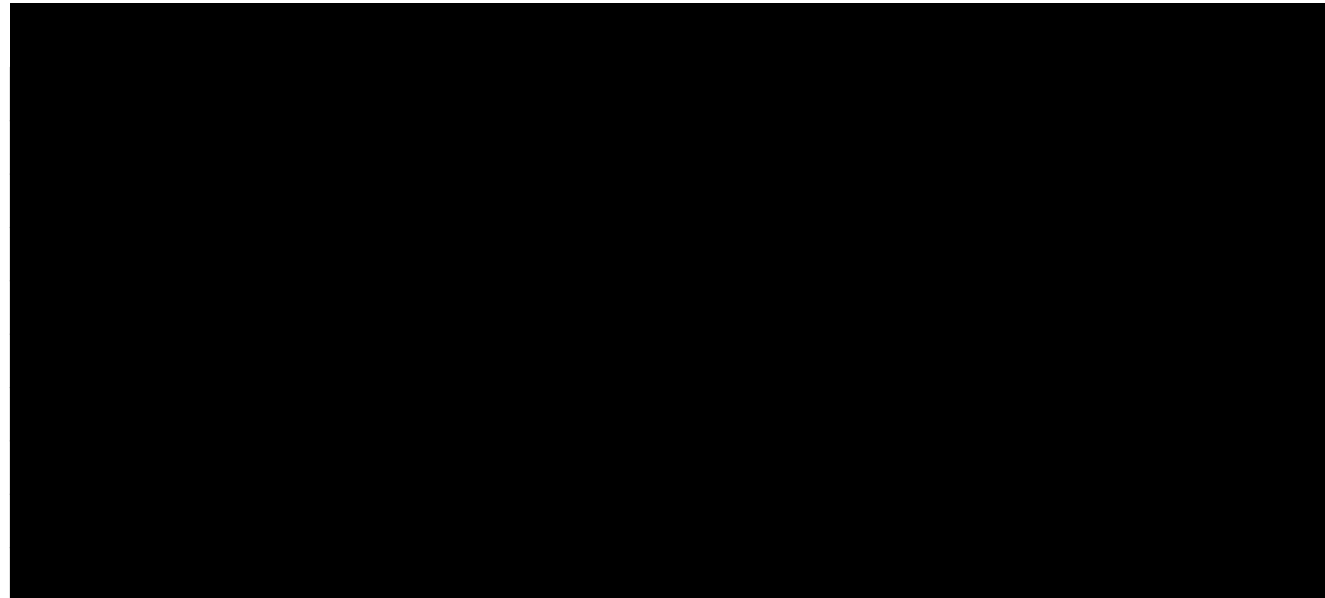
- Current role (title, company, department, work style, focus areas)
- Career goals by horizon (1yr / 3yr / 5yr / 10yr)
- Progression timeline, development skills and courses
- Professional wins log
- Network contacts with last-contact tracking

3.18 Other Tabs

- Recipes — with add-to-shopping-list integration
- Countries — visited with traveller attribution
- Bucket List — dream destinations with star rating
- Resources — reference documents VERA can read during chat

4.1 How Chat Works

Each chat message is routed through an intent-detection layer that identifies which domain the user is asking about. Only relevant context is loaded from Sheets before calling Claude — keeping prompts lean and responses accurate.



4.2 Action Execution System

VERA embeds ACTION lines in its responses. These are parsed and executed by the backend before the reply is returned to the user. The user sees a clean natural-language response — the actions have already fired.

Action format:



4.3 Full Chat Action Reference

The following 40+ actions are available for execution via chat:

Task Management



Flag Management



Goals & Projects

Calendar

Travel — Itinerary

Travel — Packing

Travel — Gym

Countries & Bucket List

Finance

Health & Wellness

People & Relationships

Home Items & Recipes

Interests & Ideas

Career

Shopping

Credit Cards & Loyalty

Takeout

Pantry & Purchases

Resources

5.1 Nightly Flag Generation

Claude reads the full life context (calendar, tasks, goals, finance, PTO, fitness, travel, interests) and generates a prioritised, deduplicated set of flags. Each flag carries:

- Source — which engine generated it
- Urgency — High / Medium / Low
- Flag summary — one concise line
- Reason — detailed explanation plus suggested action
- Key — unique identifier used for deduplication

5.2 Flag Escalation

- Day 3 unacknowledged !' urgency bumped one tier (Low!'Medium, Medium!'High)
- Day 7 unacknowledged !' "[Stale: open for 7+ days — needs attention]" appended to reason

5.3 Important Dates Engine

- 30 days before !' Low flag
- 7 days before !' Medium flag + Claude generates 3 personalised gift suggestions drawn from the Shared Interest Ledger, with estimated costs
- 1 day before !' High flag
- Last Actioned Year column prevents re-flagging for the same year
- Auto-imports birthdays from "Joint Chaos" calendar nightly

5.4 PTO Optimisation — 3-2-1 Framework

Automatically identifies optimal leave windows:

- Long Weekend — 3–4 clear workdays adjacent to weekend
- Mid-Size Week — 5–7 clear workdays
- Big Pivot — 8+ clear workdays

VERA creates soft Vera Calendar suggestions for each identified window. If a suggestion is deleted from the calendar, it is marked declined and never re-suggested. Buffer days idle for 21+ days trigger a yellow alert.

5.5 Fitness Consistency Tracker

- Weekly target configurable (default 4 sessions)
- Wednesday: Low flag if below target
- Thursday: Medium flag if zero sessions that week
- Saturday: High flag if still zero sessions
- 3+ consecutive weeks below target !' Medium flag
- Travel gap detection !' Low flag if a trip has no gym items in packing list

5.6 Financial Goal Health Check

Computes months remaining to each financial goal, required vs. current monthly contribution, and the gap. Raises a Medium or High flag if the gap is significant.

5.7 Signal Learning

Flags that remain unacknowledged for 30+ days are recorded in the Signal Learning tab. These suppressed patterns are fed back to Claude as noise filters on subsequent nightly runs, reducing repeated irrelevant flags.

5.8 Anticipator Engine (Hourly)

Scans upcoming calendar events and creates contextual prep reminders in Vera Calendar — with cooldown logic to avoid duplicating reminders already created.

5.9 Explorer Engine (Nightly)

Generates a daily discovery bulletin based on current context: active trips, flag patterns, and shared interests. Cooldown is managed via the Reminders Memory tab.

5.10 Pre-Trip Briefing (48h Before Departure)

Generates a flag summarising:

- Packing completion percentage
- Upcoming itinerary for the trip
- Flight details and status
- Hotel check-in information

5.11 Post-Trip Capture (1 Day After Return)

Sends a reflection prompt asking: Best restaurants? Trip highlights? What would you do differently? What to add to the bucket list?

5.12 Travel Day Briefing Email

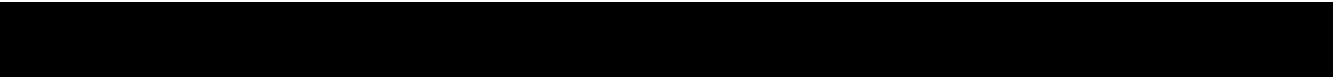
A clean morning-of-departure email containing the full day's itinerary, flight details, and weather. Designed to be shareable with travel companions.

5.13 Web Search

Integrates Serper.dev (default) or Tavily as a fallback. PII is automatically stripped from all queries. Results are synthesised naturally into the chat response.

5.14 Resource Document Intelligence

Users add Google Docs to the Resources tab. During chat, VERA fetches the document via the Drive API and makes a second Claude call with the full document text injected into context. PDFs are auto-converted via Google Drive's OCR API before processing.



6.1 Google Sheets as Database

A single Google Sheet with 50+ named tabs serves as the entire data layer. Fixed header rows provide schema consistency. The Config tab drives all system behaviour — no code changes are needed to adjust thresholds, targets, or feature flags.

6.2 Complete Tab Reference

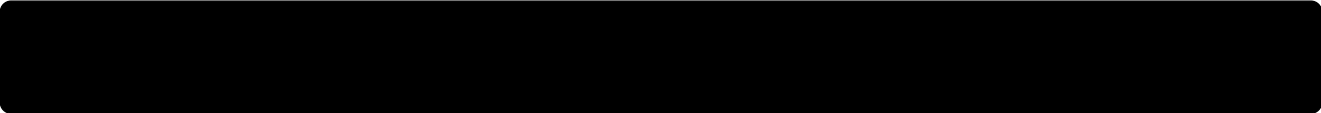
CORE



FINANCE



TRAVEL



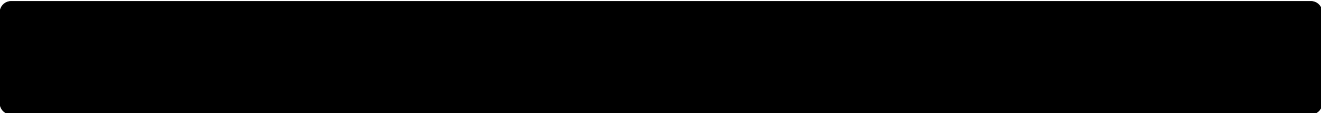
PEOPLE



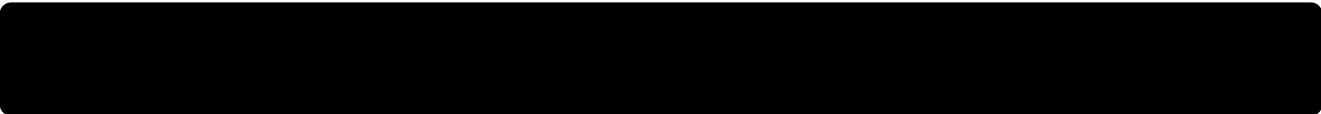
HEALTH



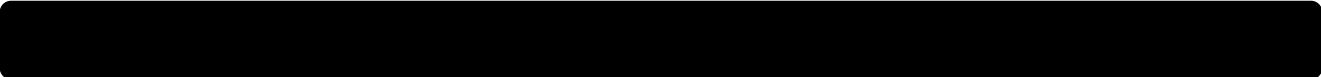
HOME



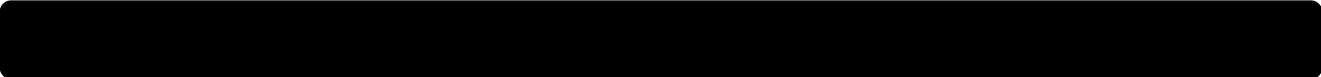
CAREER



PTO



INTELLIGENCE



6.3 Config-Driven Behaviour

The following behaviours are configurable via the Config tab without any code changes:

calendar_days_ahead	How many days of calendar to include in nightly context
task_age_threshold_days	Days before a task is considered neglected
pto_vacation_days	Ahmed's annual vacation day allowance
victoria_pto_vacation_days	Victoria's annual vacation day allowance
fitness_weekly_target	Target gym sessions per week (default: 4)
fitness_low_flag_day	Day of week to fire Low fitness flag
pretrip_briefing_hours	Hours before departure to fire pre-trip briefing
posttrip_capture_delay_days	Days after return to fire post-trip capture prompt

dates_default_lead_time

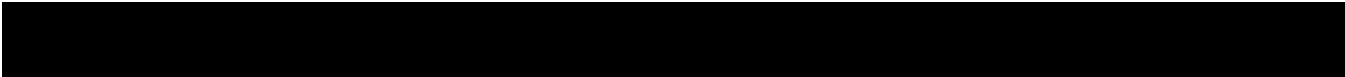
dates_medium_urgency_days	Default lead time (days) for Important Date flags
dates_high_urgency_days	Days-out threshold for Medium urgency date flags
monthly_disposable_income	Days-out threshold for High urgency date flags
	Used in financial health checks and what-if scenarios

6.4 Deduplication System

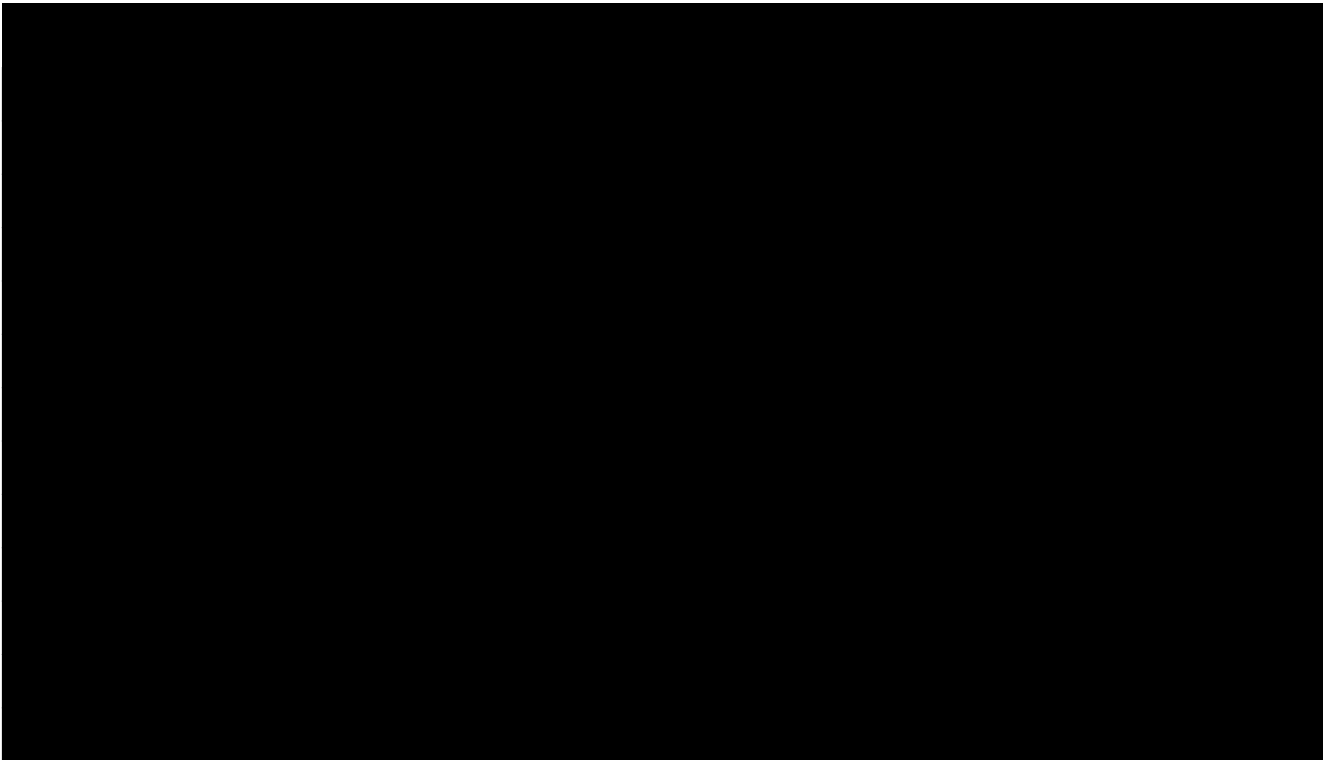
Every flag carries a unique key. The Flags tab is scanned before writing any new flag — if the key already exists in an unresolved state, the flag is silently skipped. This ensures VERA never nags about the same issue twice.

Example keys:





VERA integrates with 13 external services. All credentials are stored as Script Properties and never appear in code.



Nightly at 11 PM — nightlyRun()

1. Write Summaries + Metrics snapshot
2. Sync birthdays from "Joint Chaos" calendar to Important Dates tab
3. Fire 30/7/1-day Important Date flags (Claude generates gift suggestions at 7-day mark)
4. Write PTO snapshot; suggest Vera Calendar events for clear windows
5. Run Explorer engine (daily discovery bulletin)
6. Suggest due dates for undated tasks via Claude
7. Check financial goal health; write at-risk flags
8. Check fitness consistency (weekly target, zero-session thresholds, streak)
9. Check fitness travel gap (trips with no gym plan)
10. Check pre-trip briefings (48h before departure)
11. Check post-trip capture (1 day after return)
12. Auto-restock flagging from purchase history
13. Reset morning routine checklist for new day
14. Generate nightly flags via Claude
15. Escalate aged flags (Day 3: bump urgency; Day 7: stale marker)
16. Signal learning — record expired/unactioned flags

Daily at 7 AM — morningNudge()

- Send morning email with flags grouped High !' Medium !' Low
- Check for travel day briefings and send if departure day

Every Hour — hourlyCheck()

- Anticipator engine: scan upcoming calendar events for prep reminders
- Create Vera Calendar memos with cooldown logic

Every 15 Minutes — checkFlightStatuses_()

- 6–24h before departure: poll every 3 hours
- 1–6h before departure: poll every 60 minutes
- Under 1h before departure: poll every 15 minutes

Every 30 Minutes — runEmailScan_()

- Scan inbox for confirmation emails
- Extract structured data to enrich itinerary items

9.1 Why VERA is Different

No subscriptions, no vendor lock-in

Google Apps Script (free tier) + your own Anthropic API key. You own every byte.

Single source of truth

Everything lives in one Google Sheet. No syncing between apps, no data silos.

Claude-native intelligence

A flag isn't just "task overdue" — it's "Your dentist task has been open for 12 days while you have a dentist appointment next Tuesday."

Config-driven, not code-driven

Changing behaviour is a Config tab edit, not a code change.

Intent-aware context loading

Detects what the message is about and loads only relevant data — keeping prompts lean and responses accurate.

Real actions, not suggestions

When VERA says "I've added that to your itinerary" — it actually has. No confirmation step, no copy-paste.

Personal, not generic

Knows Ahmed's interest in jazz, Victoria's love of pottery, their upcoming Alaska cruise, Ahmed's gym target, their mortgage payoff goal.

9.2 Design Principles

1. No build pipeline

Single HTML file using Babel standalone. Deployable to GitHub Pages in seconds.

2. Fail silently on non-critical jobs

Every nightly job is wrapped in try/catch. One failing job never blocks the others.

3. Dedup everything

Every flag, calendar suggestion, and reminder carries a unique key. The system never nags twice.

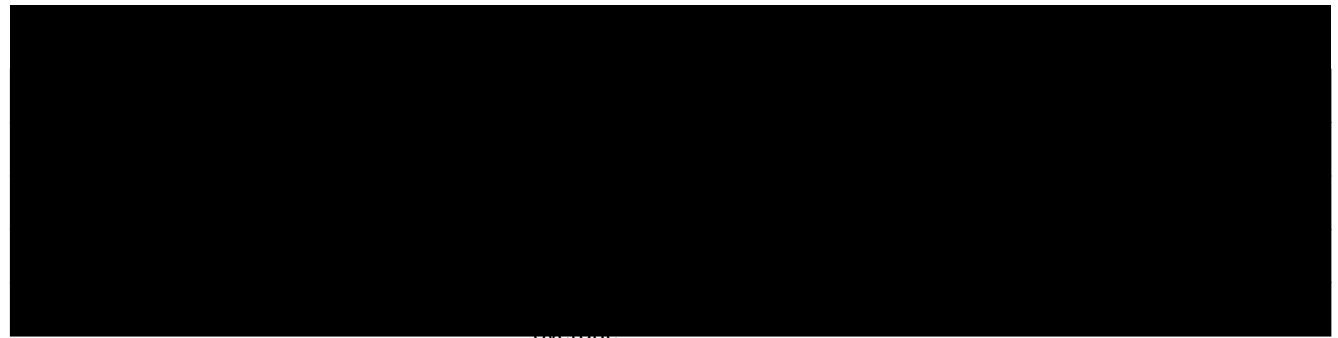
4. Google Sheets as the API

Any tool that can read Google Sheets can integrate with VERA.

5. Progressive enhancement

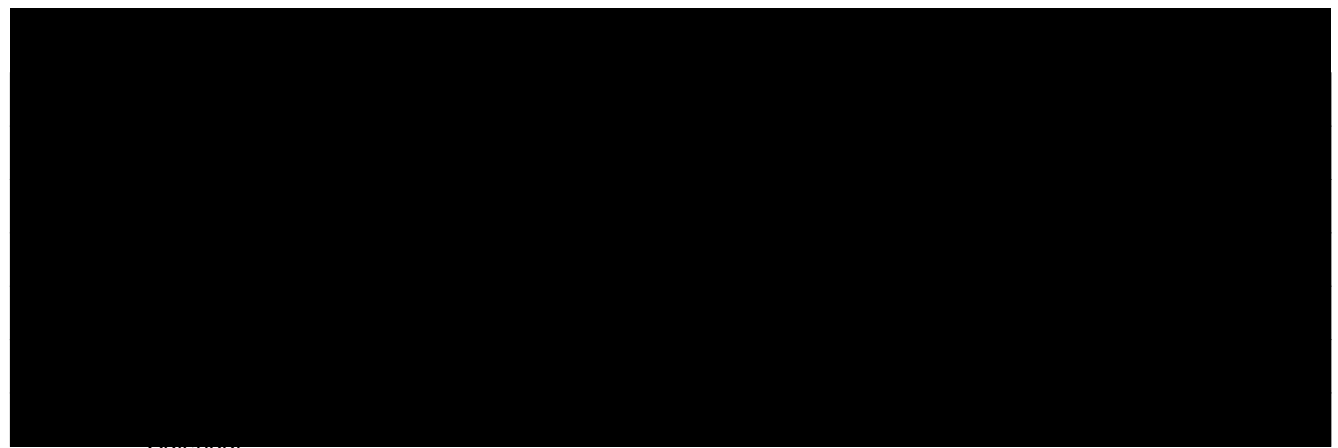
Features are gated by Config flags. Off by default until configured.

P2 — High Priority



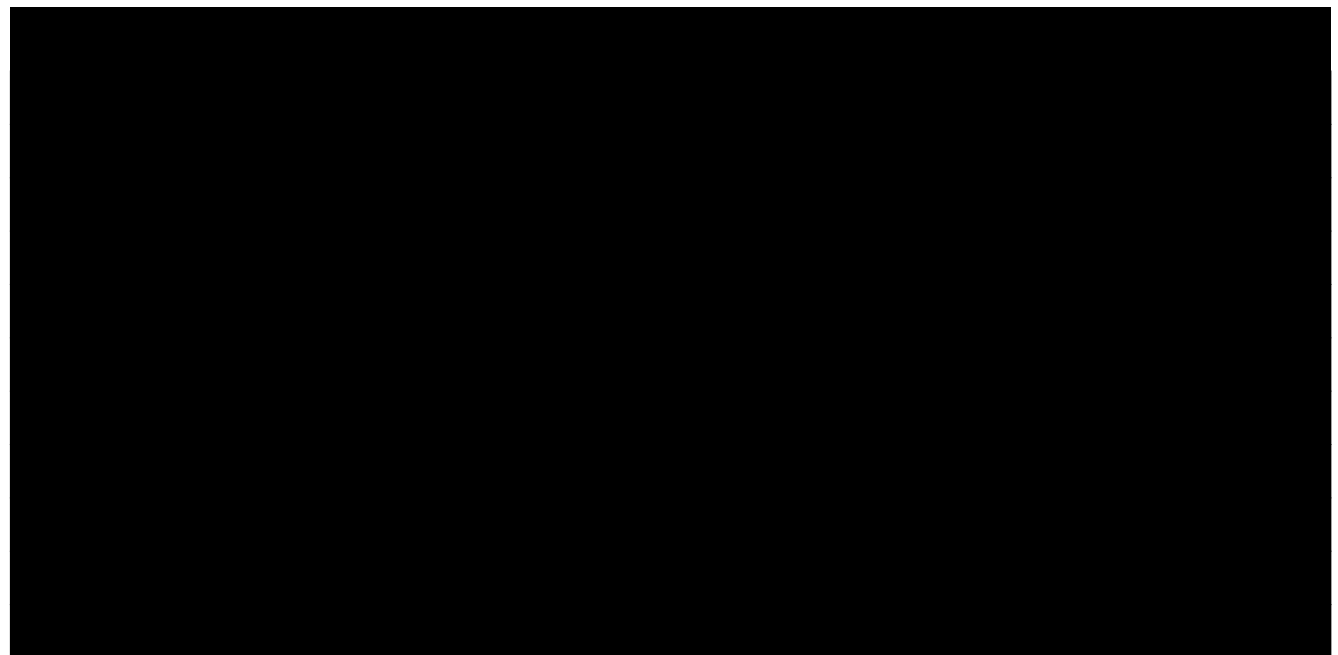
Overview:

P3 — Future



Calendar

Feature Enhancements



Prerequisites

- Google account with Google Apps Script access
- Anthropic API key (claude.ai/settings)
- Optional: AviationStack, OpenWeatherMap, Serper.dev API keys

Deployment Steps

1. Clone the repository

```
git clone https://github.com/anthropic/claude-ai-examples.git
```

2. Install clasp and log in

```
npm install -g clasp
```

3. Create the Apps Script project and push code

```
clasp create project --name "Claude AI Examples"
```

4. Set Script Properties

```
clasp set properties --key "ANTHROPIC_API_KEY"
```

5. Initialise all Sheet tabs

```
clasp run --sheet
```

6. Deploy as Web App

```
clasp deploy --webapp
```

7. Configure the dashboard

```
clasp dashboard
```

8. Enable GitHub Pages

```
clasp ghpages
```

9. Set up time triggers

```
clasp triggers
```

10. Seed Config values

```
clasp seed
```

Required Properties

Optional Properties

[REDACTED]

[REDACTED]